- ...nsists of seven questions. Please explain your answers step by step.
- Write **four** digits after the comma/point.
- Use **radian** when calculating with **trigonometric** functions.

...inish at **15:30**. The total time of the exam is **90** minutes.

Good luck!

QUESTIONS

Q1. Consider the function $f(x) = e^x$.

a. Determine the Taylor series expansion about $x_0 = 0$ with the first four terms. (12 pts)

b. Calculate the approximate value of the e^{-2} in the expansion you obtained and find the true relati... (8 pts)

Answer:

Q2. This partial code implements the bisection method for finding a root of the nonlinear equation $f(x) = 0$.

```
while( abs(x0-x1) > tol ) % absolute error
tolerance
    x2 = x1 - f(x1)*(x1-x0)/(f(x1)-f(x0));
    if f(x2)*f(x1) > 0
        ... 1 ...
    else
        ... 2 ...
    end
end
```

The Matlab function f computes the function values and you can assume that you start with an interval [a, b] satisfying $f(a).f(b) < 0$. Select the suitable replacement code for the blanks marked 1 and 2 from the list below. (5 pts)

A) **1:** b=mid; fb=fmid; **2:** a=mid; fa=fmid;

B) **1:** a=mid; fa=fmid; **2:** b=mid; fb=fmid;

C) **1:** a=mid; fb=fmid; **2:** b=mid; fa=fmid;

D) **1:** a=mid; **2:** b=mid;

Q3. This partial code implements the method of Regula- Falsi method for finding a root of the nonlinear equation $f(x) = 0$, where the Matlab function f computes the function value. Provide suitable code to replace the blanks marked 1 and 2. (5 pts)

```
while( abs(x0-x1) > tol ) % absolute error
tolerance
    x2 = x1 - f(x1)*(x1-x0)/(f(x1)-f(x0));
    if f(x2)*f(x1) > 0
        ... 1 ...
    else
        ... 2 ...
    end
end
```

A) $1: x_1 = x_2;\ \ 2: x_0 = x_2;$

B) $1: x_2 = x_1;\ \ 2: x_2 = x_0;$

C) $1: x_0 = x_2;\ \ 2: x_1 = x_2;$

D) $1: x_2 = x_0;\ \ 2: x_2 = x_1;$

Q4. Find the fourth root of 200 by finding the numerical solution of the equation $x^4 - 200 = 0$. Use Newton's method. Start at $x = 8$ and carry out the first three iterations. (20 pts)

Answer:

Q5. The equation $f(x) = \cos(x) + 2\sin(x) + x^2 = 0$ is given. Find an approximate root of this equation using Secant method. (Initial points are $x_0 = 0$ *and* $x_1 = -0.1$.) (15 pts)

Answer:

Q6. Consider the fixed point iteration $x_{k+1} = g(x_k)$ with $g(x) = \frac{x}{3} + \frac{4}{3x}$. Write a root finding problem suitable for this function. (12 pts)

Answer: